

1 de-Rham Cohomology

In this section, I will explain the study of “holes” on a manifold using algebraic methods. We know that an exact form, $\omega = d\alpha$ is also closed by virtue of $d^2 = 0$. Then we ask the question, is the converse true? That is, if a form ω is closed ($d\omega = 0$), then can we write this form $\omega = d(\text{something})$? The answer to this question is de-Rham cohomology theory. To understand how answering this question will give information about the “holes” on the manifold, let me first remove the “co” in cohomology.

1.1 Homology

Consider a sub-manifold,

$$C \subset M \tag{1.1}$$

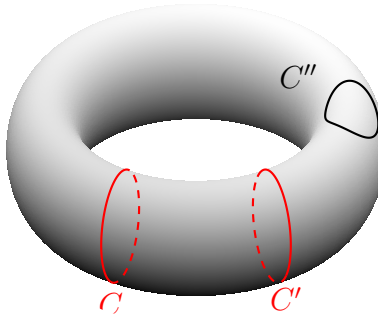
Let us assume that C itself has no boundary. We write this as,

$$\partial C = 0 \tag{1.2}$$

The question we ask is : is C a boundary of some other sub-manifold $D \subset M$? In symbols,

$$\partial D \stackrel{?}{=} C \tag{1.3}$$

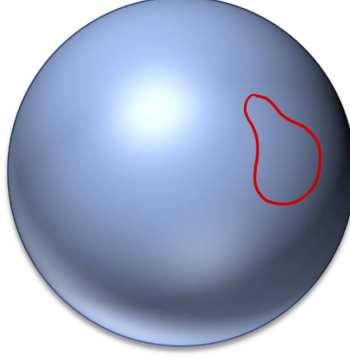
Consider the picture below¹,



Here, the sub-manifold C'' bounds a region, while C and C' does not. Now, if we look at²,

¹code is taken from [here](#)

²pictures from David Tong notes on Supersymmetry



Here, no matter how we draw a loop, it will always bound some region on the sphere.

This is the key point of homology theory. We can differentiate between - in this case- a sphere and a torus by looking at loops (sub manifolds which have no boundary) that do not bound something else. To formalise this a bit more, we define the following sets,

$$Z_p(M) := \left\{ C \subset M \mid \dim C = p \quad \text{and} \quad \partial C = 0 \right\} \quad (1.4)$$

and,

$$B_p(M) := \left\{ C \subset M \mid \dim D = p \quad \text{and} \quad \exists D \subset M : \dim D = p + 1 \quad \text{and} \quad C = \partial D \right\} \quad (1.5)$$

The first set contains all the p -dimensional sub-manifolds of M that have no boundary. The second contains, all the p -dimensional sub-manifolds of M which bounds another $p + 1$ -dimensional sub-manifold of M . Since a boundary of a boundary is zero, it is clear immediately that,

$$B_p(M) \subseteq Z_p(M) \quad (1.6)$$

As I sketched above, we are interested in things with no boundary that do not bound anything. That means, we have to somehow “take out” B from Z . We will do this by means of an equivalence class. We define,

$$\text{for } z, z' \in Z_p(M) : z \sim z' \iff z - z' \in B_p(M) \quad (1.7)$$

Then, define the p th homology group as the quotient of $Z_p(M)$ with this equivalence class,

$$H_p(M) := Z_p(M) / B_p(M) \quad (1.8)$$

Example 1 : The Torus

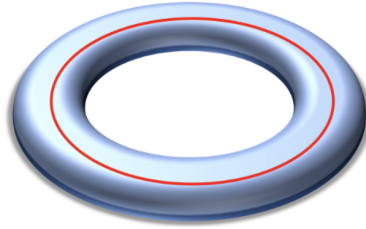
As an exercise, let us figure out the homology classes of the torus. First, $H_0(T^2)$. Zero dimensional sub-manifold is a point, and the point trivially has no boundary. Also, the point cannot bound anything because there are no -1 dimensional sub-manifold. Now, take two points, p_0 and p_1 . The difference can be thought of as a boundary of a curve connecting p_0 to p_1 . Hence, all points are equivalent. This gives us,

$$\dim H_0(T^2) = 1 \iff H_0(T^2) \cong (\mathbb{Z}, +) \quad (1.9)$$

What the right hand side says is that $H_0(M)$ is isomorphic to a group generated by one element, which is $(\mathbb{Z}, +)$.

This calculation also tells us something more. For any manifold which is simply connected (i.e, exists a path between any two points), the zeroth homology group is the same! This actually is true, and the formal proof almost looks like what we did. Also, if the manifold has say N connected components, then the dimensions of $H_0(M)$ will be just N , corresponding to N points, each from one of the connected components.

Now, let us find $H_1(M)$. As we saw before, C and C' are one dimensional sub-manifolds with no boundary that are themselves not boundaries, while C'' is only a member in $Z_1(M)$. We would say that C and C' are equivalent because the difference bounds the region between the loops. Now, there is another independent loop. This is shown below,



A little thought can convince you that there are no more such one dimensional sub-manifolds. Hence,

$$\dim H_1(T^2) = 2 \iff H_1(T^2) \cong (\mathbb{Z}, +) \oplus (\mathbb{Z}, +) \quad (1.10)$$

Now, for $H_2(T^2)$, we have the entire surface of the torus, which has no boundary, as well as not bounding any other sub-manifold. Hence,

$$\dim H_2(T^2) = 1 \iff H_2(T^2) \cong (\mathbb{Z}, +) \quad (1.11)$$

Example 2 : The Sphere

You can convince yourself that for the sphere, the homology groups are,

$$H_0(S^2) \cong H_2(S^2) \cong (\mathbb{Z}, +) \quad (1.12)$$

and,

$$H_1(S^2) \cong \{0\} \quad (1.13)$$

Notice that the homology groups differ for S^2 and T^2 . The homology groups (rather intuitively) are topological invariants. Therefore, having different homology groups immediately tells us that T^2 and S^2 are topologically different. But remember, having the same homology groups for two manifolds does not guarantee that there is a homeomorphism between them, just like for any other topological invariant.

1.2 de Rham Cohomology

We will now discuss differential forms, and a similar construction like the homology groups. This will be called de Rham cohomology, and indeed it will turn out to be the cohomology is the dual vector space of the homology group. Let us jump right into it. We work with p -forms, $\Omega^p(M)$, and define the following subsets,

$$Z^p(M) := \left\{ \omega \in \Omega^p(M) \mid d\omega = 0 \right\} \quad (1.14)$$

and,

$$B^p(M) := \left\{ \omega \in \Omega^p(M) \mid \exists \eta \in \Omega^{p-1}(M) : \omega = d\eta \right\} \quad (1.15)$$

Because of $d^2 = 0$, we have,

$$B^p(M) \subseteq Z^p(M) \quad (1.16)$$

The similarity is already apparent. It is stemming from $\partial^2 = 0$, as well as $d^2 = 0$. Here are some quick remarks,

- If $\omega \in Z^p(M)$ and $\eta \in Z^q(M)$ then $\omega \wedge \eta \in Z^{p+q}(M)$.
The proof is very easy, all we have to do is to use the graded Leibniz rule to figure out that $\omega \wedge \eta$ is closed.
- If $\omega \in Z^p(M)$ and $\eta \in B^q(M)$ then $\omega \wedge \eta \in B^{p+q}(M)$.
To prove this, we just have to notice that $\omega \wedge \eta = \omega \wedge d\phi = d(\omega \wedge \phi)$

- If $\omega \in B^p(M)$ and $\eta \in B^q(M)$ then $\omega \wedge \eta \in B^{p+q}(M)$.

This is proved in a similar way, $\omega \wedge \eta = \omega \wedge d\phi = d(\omega \wedge \phi)$. In this case, there is one more (at least) form that when differentiated gives $\omega \wedge \eta$.

Now, we find the cohomology, i.e, we look for elements in Z that are not in B . i.e, closed p -forms that are not exact. We know how to get to it. We first define the equivalence relation,

$$\text{for } \omega, \omega' \in Z^p(M) : \omega \sim \omega' \iff \omega - \omega' \in B^p(M) \quad (1.17)$$

Then, we define the cohomology group³ as the quotient set,

$$H^p(M) = Z^p(M)/B^p(M) \quad (1.18)$$

Example 1 : $H^0(M)$

Let us understand the simplest case, $H^0(M)$. It is obvious that none of the zero forms are exact, because there are no -1 -forms. Now, for a zero form to be closed, we need it to be a constant function. Hence, we have one generator for $H^0(M)$. As a vector space,

$$H^0(M) \cong \mathbb{R} \quad (1.19)$$

If there were N connected components, then the dimensions of $H^0(M)$ would be N .

Example 2 : $H^1(\mathbb{R})$

Now, let us look at some examples to compute $H^1(M)$. First, let $M = \mathbb{R}$. Since the manifold is one dimensional, every one form is closed. Now, we ask the exactness. Let,

$$\omega \in \Omega^1\mathbb{R} : \omega(x) = f(x)dx \quad (1.20)$$

We can define a zero form,

$$F(x) = \int_0^x dy f(y) \quad (1.21)$$

Then by fundamental theorem,

$$\frac{dF(x)}{dx} = f(x) \implies dF = f(x)dx = \omega \quad (1.22)$$

Hence, all one forms are exact. This means,

$$H^1(\mathbb{R}) = \{0\} \quad (1.23)$$

³actually a vector space for this specific case

Example 3 : $H^1(S^1)$

Let us take $M = S^1$ next. Consider a one form,

$$\omega = f(\theta)d\theta \quad (1.24)$$

As before, define,

$$F(\theta) = \int_0^\theta d\theta' f(\theta') \quad (1.25)$$

But now, since we are on the circle, to have well definition, F should be periodic. i.e, $F(2\pi) = F(0)$. Since $F(0) = 0$, we must have,

$$F(2\pi) = \int_0^{2\pi} d\theta' f(\theta') = 0 \quad (1.26)$$

We must satisfy this condition, only then we can write $\omega = dF$. Otherwise, we do not have a well defined F . Now, if this condition is not met, then the form ω is not exact. Let ω and ω' be inexact one forms on the circle. Then,

$$\int_0^{2\pi} \omega \neq 0 \quad (1.27)$$

We can then define a number,

$$a = \frac{\int_0^{2\pi} \omega'}{\int_0^{2\pi} \omega} \quad (1.28)$$

Then,

$$\int_0^{2\pi} (\omega' - a\omega) = 0 \implies \omega' - a\omega \in B^1(S^1) \quad (1.29)$$

Therefore, given a closed but not exact one form ω , every other closed but not exact one form ω' is cohomologous to $a\omega$ for $a \in \mathbb{R}$. Which means there is only one independent closed but not exact one form, and,

$$\dim H^1(S^1) = 1 \quad (1.30)$$

Example 4 : $H^1(\mathbb{R}^2 - \{(0,0)\})$

We take a lesson from this example. On a circle, if a one form is exact, $\omega = dF$, then,

$$\int_{S^1} \omega = \int_{S^1} dF = \int_{\partial S^1} F = 0 \quad (1.31)$$

The last equality follows because the circle has no boundary. Therefore, if we can find a one form that has,

$$\int_{S^1} \omega \neq 0 \quad (1.32)$$

then we immediately know that ω is not exact. We will use this idea for any general manifold. We will embed a circle in the manifold and look for one forms that when integrated over this circle to find closed but not exact one forms. The stereotypical example is,

$$M = \mathbb{R}^2 - \{(0, 0)\} \quad \omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \quad (1.33)$$

Showing the closedness is easy. To show that this is not exact, we do the integral on a sub-manifold, a unit circle. On this sub-manifold, we can parameterise the coordinates as,

$$x = \cos \theta \quad \text{and} \quad y = \sin \theta \quad (1.34)$$

Then, the forms pulled back to the circle using this parameterisation map are,

$$dx \rightarrow -\sin \theta d\theta \quad \text{and} \quad dy \rightarrow \cos \theta d\theta \quad (1.35)$$

Then,

$$\int_{S^1} \omega = \int_0^{2\pi} (\sin^2 \theta d\theta + \cos^2 \theta d\theta) = 2\pi \neq 0 \quad (1.36)$$

This form is not exact. However, it is easy to check that,

$$\omega = dF, \quad \text{where } F(x, y) = \tan^{-1}(y/x) \quad (1.37)$$

What happened? The answer is that the function is defined on $\mathbb{R}^2 - \{y\text{-axis}\}$, not on our manifold M . If this was our manifold, then there is no way to embed a circle such that it encircles the origin. The manifold will have two connected parts, the left and right half planes. In both these planes, the one form will be exact⁴.

This trick can be generalised further. If we have $\omega = d\eta$ an exact p -form, then embed S a compact p dimensional manifold with no boundary on M . Let $\phi : S \rightarrow M$ be the embedding. Then we have,

$$\int_S \phi^* \omega = \int_S \phi^* (d\eta) = \int_S d(\phi^* \eta) = \int_{\partial S} \phi^* \eta = 0 \quad (1.38)$$

Therefore, we can play the same trick for p -forms. Embed a compact boundary-less p -dimensional sub-manifold. Then if a p -form pulled back to this sub-manifold using the embedding has non vanishing integral, it is not exact.

⁴As a quick exercise, try finding the integral around a loop at say $(1, 1)$ and confirm that this is zero.

1.3 de Rham's Theorem

Let me now describe the connection between homology and cohomology. This goes by the name of de Rham's theorem. First, we show that the homology classes and the cohomology classes are dual to each other, hence isomorphic. The homology classes consists of sub-manifolds of M , and the cohomology consists of forms. Given a p -dimensional sub-manifold $C \subset M$ and a p -form ω , we can define an inner-product between these, i.e, a map to the reals,

$$(C, \omega) := \int_C \omega \quad (1.39)$$

The inner-product can be defined to be zero when the dimension of the manifold C and the degree of the form do not match. Using this, we can define an inner-product between the homology class and the cohomology class as,

$$\Lambda([C], [\omega]) = (C, \omega) \quad (1.40)$$

This is well defined, independent of the representatives. This follows from Stoke's theorem.

- The inner-product well defined for the homology class

proof : If $C_1 - C_2 = \partial D$, then for a closed form ω

$$\int_{C_1} \omega - \int_{C_2} \omega = \int_{\partial D} \omega = \int_D d\omega = 0 \quad (1.41)$$

- The inner-product well defined for the cohomology class

proof : If $\omega - \omega' = d\eta$, then for a closed sub-manifold C ⁵

$$\int_C (\omega - \omega') = \int_C d\eta = \int_{\partial C} \eta = 0 \quad (1.42)$$

Remark : In this language the Stokes' theorem takes the cute form,

$$(C, d\omega) = (\partial C, \omega) \quad (1.43)$$

Now that we have the inner-product, we will have to show that the dual space of homology with respect to the inner-product is the cohomology. This is the de Rham's theorem and I am not proving it here. (I do not know the proof myself!) This theorem is what allows us to study the forms and infer properties about the manifold.

⁵meaning no boundary, $\partial C = 0$

1.4 Applications to Physics : Scalar Potential in Electrostatics

Let us discuss the concepts so far in the context of physics. When we learned electrostatics, we assumed that the electric field is the gradient of a scalar. Now, we are wiser. We know that the electric field is a one form,

$$E(x) = E_\mu(x)dx^\mu \quad (1.44)$$

The Maxwell equation when we are in static situations is,

$$dE = 0 \quad (1.45)$$

That is, electric field is closed. Now, the question we ask is, is it also exact? Only then do we have a scalar potential $\phi(x)$ such that,

$$E = -d\phi \quad (1.46)$$

By now, we know the answer. Embed a circle into the manifold. If the integral of E on the circle vanishes, then we know that the electric field is exact. We saw an example for the manifold with one point removed, there were non exact forms. We remarked there that if the manifold is simply connected, then all one forms are exact. This is the answer. In electrostatics, we usually consider manifolds that are simply connected. Hence, we always can write down a scalar potential.

1.5 Applications to Physics : The Aharanov-Bohm Effect

Let us first do a warm up problem. Finding the magnetic field due to an infinitely long cylindrical wire. For convenience, we will be using cylindrical coordinates (r, θ, z) . z is a well defined coordinate, and dz is hence a well defined one form all over $\mathbb{R}^3$. dr however, is defined only away from the $r = 0$ line. The θ coordinate is defined only modulo 2π . We can define a one-form,

$$d\theta = \frac{-y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy \quad (1.47)$$

We know that this one form is inexact, because we have removed the $r = 0$ line from the manifold. This is the reason for the appearance of d . Let us assume that the current is flowing in the z direction, symmetrically.

$$j = f(r)dz \quad (1.48)$$

To use it in the Maxwell equation, we need to find the dual. In cartesian, we know,

$$\star dz = dx \wedge dy \quad (1.49)$$

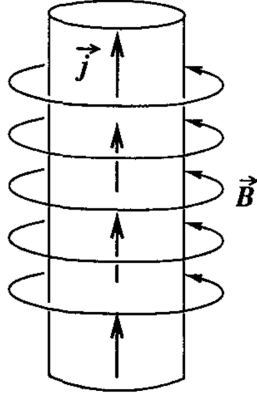
Then, away from the z -axis, we can write,

$$dx = \cos \theta dr - r \sin \theta d\theta \quad \text{and} \quad dy = \sin \theta dr + r \cos \theta d\theta \quad (1.50)$$

Then,

$$\star dz = r dr \wedge d\theta \quad (1.51)$$

Now, if we use our three dimensional intuition, we know the magnetic field(vector) should be winding the wire anti-clockwise, something like this⁶,



That is,

$$\star B = g(r) d\theta \quad (1.52)$$

Taking duals, and using,

$$\star d\theta = \frac{1}{r} dz \wedge dr \quad (1.53)$$

we get,

$$B = \frac{g(r)}{r} dz \wedge dr \quad (1.54)$$

It is then clear that $dB = 0$, because the function only depends on r and if we differentiate with respect to r and add a $dr \wedge$, we will get zero. The second Maxwell equation is,

$$d \star B = \star j \quad (1.55)$$

This becomes,

$$\frac{dg(r)}{dr} = r f(r) \quad (1.56)$$

⁶figure from John Baez's book

The formal solution is,

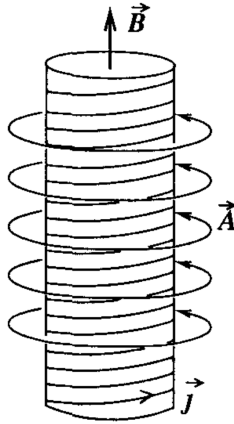
$$g(r) = g(0) + \int_0^r ds s f(s) \quad (1.57)$$

For the magnetic field to be well behaved as $r \rightarrow 0$, we need to choose $g(0) = 0$. There is still some ambiguity in the solution. If we find a 2-form C such that,

$$dC = 0 = d \star C \quad (1.58)$$

then we can add this to the solved B , and nothing would change. A two form with constant coefficients will satisfy these properties. But if we demand that the magnetic field goes to zero at infinity, then we are forced to choose $C = 0$. Then, we have a unique solution to the problem.

Now we come to the actual problem. We wind the a current carrying wire around a cylinder of radius R , like this,



For an infinitely long solenoid, we get a constant magnetic field along the z -axis as shown above, and zero field outside. i.e,

$$\star B = f(r) dz \implies B = f(r) r dr \wedge d\theta \quad (1.59)$$

where $f(r)$ is constant for $r < R$ and zero otherwise. Now we ask the question, what is the corresponding vector potential? This is exactly the same as the warm up example, and we borrow the answer,

$$A = g(r) d\theta = \left(\int_0^r ds s f(s) \right) d\theta \quad (1.60)$$

We see that, we can simply write,

$$A = \frac{\Phi}{2\pi} d\theta \quad (1.61)$$

where,

$$\Phi = \left(\int_0^r ds s f(s) \right) \quad (1.62)$$

Φ is the magnetic flux, the integral of the magnetic field over a disc D . To see this, we just have to do,

$$\oint_{\partial D} A = \int_D dA = \int_D B \quad (1.63)$$

and,

$$\frac{\Phi}{2\pi} \oint_{\partial D} d\theta = \Phi \quad (1.64)$$

Of course, this vector potential is not unique, but it does not matter to us since we are looking for some vector potential.

Now, we let a charged particle move outside the solenoid. Here, there is no magnetic field. Classically we expect the charged particle to be unaffected. But quantum mechanics tells another story. Let the particle start out with the wave function $\psi(a)$ at $a \in M$ at time $t = 0$. Then we wish to calculate the transition amplitude that the particle will have the wave function $\phi(b)$ after some time T . Denote the set of all paths from a to b in time T as,

$$\mathcal{P} = \left\{ \gamma : [0, T] \rightarrow M \mid \gamma(0) = a \text{ and } \gamma(T) = b \right\} \quad (1.65)$$

Then the transition amplitude is given by the path integral expression,

$$\phi(b) = \int \mathcal{D}\gamma \psi(a) \exp \left\{ \frac{i}{\hbar} \left(S_0[\gamma] - q \int_{\gamma} A \right) \right\} \quad (1.66)$$

Here, S_0 is the free particle action. In particular, if we are interested in a particle moving in a loop ($a = b$), then the additional phase factor $\exp \left\{ -\frac{i}{\hbar} q \oint_{\gamma} A \right\}$ will be different for different paths, and will result in some interference between paths. i.e, the particle's transition amplitude feels the presence of the vector potential, which does not match our classical intuition.

If we have a solenoid of radius R , then the charged particle will effectively get to see the manifold,

$$M = \mathbb{R}^3 - \{r < R\} \quad (1.67)$$

Outside the solenoid,

$$\oint_{\gamma} A = \oint_{\partial D} A = \int_D B = \Phi \quad (1.68)$$

Therefore, as the particle travels on a loop around the solenoid, it picks up a phase,

$$\exp\left\{\frac{iq\Phi}{\hbar}\right\} \tag{1.69}$$

Therefore, if we start with two particles with identical wave functions and let one of them travel in a loop around the solenoid, this will pick up a phase from the motion and we can compare the phases of the two particles and observe this phase. This is the Aharonov-Bohm effect.

The math lesson that we take away is this : We have a manifold which is not simply connected. Then, there is a closed one form⁷, $A = \frac{\Phi}{2\pi}d\theta$ whose integral on a circle embedded in the manifold does not vanish. This existence of a closed but non exact form has physically observable consequences, namely the Aharonov-Bohm phase.

⁷this is closed since outside the solenoid, Φ is constant, and we already have shown that $d(d\theta) = 0$